SkillNavora

MODULE 18 HANDBOOK

Module 18: SFMC APIs and Integrations

6 sections, 53 topics explained, plus the detailed lessons, examples, exercises and official links.

Salesforce Marketing Cloud (SFMC) Academy. Verify product names, editions, licences and certifications on Salesforce's official pages.

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Edition: November 2026. Practice labs marked "account" need a provisioned Marketing Cloud account. Other labs use sample files or paper.

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Module 18: SFMC APIs and Integrations

Level: Advanced. REST and SOAP APIs, installed packages, OAuth, and integration patterns with Salesforce and other systems.

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Detailed lessons

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18.1 REST authentication with an installed package

WHAT IS IT?

> An installed package in SFMC gives an app a client ID and secret. The app requests an access token from the account's auth endpoint, then calls REST endpoints with the token. Token requests use your account subdomain.

WHY IS IT IMPORTANT?

> Every integration starts with authentication; mistakes cause 401 errors and leaked credentials.

HOW DOES IT WORK AND THE STEPS

1. Create an installed package with the required API permissions.
2. Copy the client ID, secret and subdomain (store them in a secrets manager).
3. Request a token and call one endpoint.

REAL-WORLD EXAMPLE

> POST to the token endpoint, then GET a data extension list with the bearer token.

CODE EXAMPLE, WITH LINE-BY-LINE NOTES

```
# Token request (replace the subdomain and credentials; never commit them)
curl -X POST "https://YOUR_SUBDOMAIN.auth.marketingcloudapis.com/v2/token" \
  -H "Content-Type: application/json" \
  -d '{"grant_type":"client_credentials","client_id":"YOUR_ID","client_secret":"YOUR_SECRET"}'
# The response contains access_token. Use it as: Authorization: Bearer <access_token>
```

COMMON ERRORS

Putting the client secret in front-end code.

INTERVIEW QUESTION

Explain how OAuth works in an SFMC integration.

HANDS-ON EXERCISE

Authenticate with a sandbox account and list data extensions with Postman.

OFFICIAL DOCUMENTATION AND FURTHER READING

- Marketing Cloud REST and SOAP API docs

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18.2 REST versus SOAP and common patterns

WHAT IS IT?

> REST uses JSON over HTTP and is the recommended approach for new work. SOAP uses XML and WSDL and remains available for some legacy operations. Patterns: create triggered sends, upsert contacts, read tracking data, manage assets.

WHY IS IT IMPORTANT?

> Choosing the right API avoids rework; REST is simpler to test.

HOW DOES IT WORK AND THE STEPS

1. Use REST for new integrations.
2. Check the docs for each operation's rate limits.
3. Log every request and response status.

REAL-WORLD EXAMPLE

> Triggered send: POST with the subscriber, the email definition and personalisation attributes.

COMMON ERRORS

Sending bulk data one record at a time instead of using batch endpoints.

INTERVIEW QUESTION

REST versus SOAP: which would you use and why?

HANDS-ON EXERCISE

Describe a triggered-send integration with request, response and error handling.

OFFICIAL DOCUMENTATION AND FURTHER READING

- Salesforce Developer Documentation

18.1 API Fundamentals: every topic explained

What is an API?. A defined way for one program to ask another for data or actions.

REST APIs. APIs that use URLs, HTTP methods and JSON.

HTTP methods. GET reads, POST creates, PUT and PATCH update, DELETE removes.

JSON. JavaScript Object Notation: text data with keys and values.

Endpoints. The URL for one API resource or action.

Request and response. The client sends a request; the server returns a response with a status code and body.

Authentication. Proving who is calling, usually with a token.

OAuth 2.0. A standard for getting access tokens without sharing passwords.

API keys. A key that identifies the calling app (SFMC mainly uses OAuth tokens).

Rate limits. Limits on calls per period; respect them with backoff.

Pagination. Getting large results in pages.

Webhooks. Calls sent to your URL when events happen.

18.2 Marketing Cloud APIs: every topic explained

Marketing Cloud Engagement API architecture. REST and SOAP APIs on account-specific endpoints, authenticated through installed packages.

Installed packages. Packages created in Setup that hold API integration components and credentials.

API integration components. The API integration in a package defines the client ID, secret and scopes.

Server-to-server integration. Server-to-server integration uses the client credentials grant, without a user login.

REST API. JSON API for most modern operations.

SOAP API. XML API with WSDL for many legacy and bulk operations.

Authentication. Get a token from the auth endpoint with the package credentials.

Access tokens. Tokens expire; request a new one before expiry.

Scopes. Permissions requested for the token, such as data extension read or email send.

API permissions. Choose the minimum permissions in the package.

18.3 REST API Applications: every topic explained

Creating data extensions. Create data extensions through the API.

Retrieving data. Read rows from data extensions.

Managing assets. Create and update Content Builder assets.

Creating email sends. Start sends through the API.

Triggered sends. Send transactional or triggered messages through the API.

Journey interactions where supported. Fire API events that inject contacts into journeys, where configured.

Subscriber management. Update subscriber status and attributes.

Contact management. Create and update contacts.

18.4 SOAP API: every topic explained

SOAP fundamentals. XML messages in an envelope with a header and body.

WSDL. The service description file listing operations and objects.

Objects. Objects such as DataExtension, Subscriber and Send.

Retrieve operations. Retrieve requests with filters and properties.

Create operations. Create requests to add objects.

Update operations. Update requests to change objects.

Delete operations. Delete requests to remove objects.

SOAP request structure. Envelope, header with token, body with the request object.

18.5 Integrations: every topic explained

Salesforce Sales Cloud. Marketing Cloud Connect links Sales Cloud data and activities.

Salesforce Service Cloud. Service Cloud cases and contacts used in journeys.

Salesforce CRM. Synchronised CRM objects for segmentation and journeys.

External CRM systems. Other CRMs connect through APIs or middleware.

E-commerce platforms. Orders and carts from commerce systems for journeys.

Payment systems. Payment events such as failures trigger messages.

Web applications. Apps call the API to trigger messages or update data.

External databases. Data exchanged by files, APIs or integration tools.

18.6 API Development and Testing: every topic explained

Postman. A tool to build and test API requests.

Authentication testing. Check token requests and expiry handling.

Request testing. Send sample requests with test data.

Response validation. Check status codes and response bodies.

Error handling. Handle 4xx and 5xx responses with clear actions.

Logging. Log requests and responses without secrets.

API security. Protect credentials, use HTTPS and least privilege.